

Online parameter reconstruction of a bilinear control problem

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Summary

- 1 Motivation
- 2 Estimator construction & results
- 3 Conclusion & outlook

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MRI & Bloch-Torrey



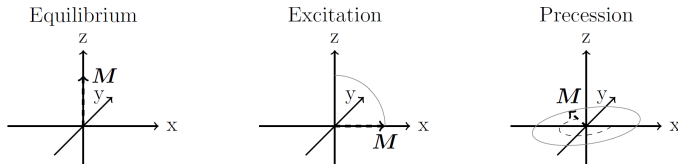
Bloch-Torrey equation:

$$\dot{z}^* + q \times z^* + \text{diag} \left(\frac{1}{T_2}, \frac{1}{T_2}, \frac{1}{T_1} \right) z^* + \Delta z^* = z_{\text{eq}}^*.$$

Observations:

$$y(t) = (\langle c_j, z^* \rangle)_{j \in \{1, \dots, n\}}.$$

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Framework & Objectives

System :

$$\begin{cases} \dot{z}^* + H_0(z^*)\theta^* + C(q)z^* = 0, \\ y = Rz^*. \end{cases} \quad (S)$$

Objectives:

- reconstruct the parameter θ^*
- do so quickly and precisely $\implies$ online method
- design a (feedback?) control q to optimise the reconstruction

Literature

Classical approach (Baumeister et al. 1997; Kügler 2010; Boiger and Kaltenbacher 2016): estimators θ and z such that

$$\begin{cases} \dot{z} + H_0(z)\theta + C(q)z = H_0(Re^z)\theta - \lambda R^* Re^z \\ \dot{\theta} = H_0(z)^* Re^z, \\ e^z = z - z^* \\ e^\theta = \theta - \theta^*. \end{cases}$$

Energy estimates:

$$\Gamma = \frac{1}{2} \left(\|e^z\|^2 + \|e^\theta\|^2 \right)$$

$$\dot{\Gamma} = -\lambda \|Re^z\|^2 + \langle H_0(Re^z)e^\theta - H_0(R^\perp e^z)\theta^*, e^z \rangle - \langle R^\perp e^z, H_0(z)e^\theta \rangle$$

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Estimator construction

$$\begin{cases} \dot{z}^* + H_0(z^*)\theta^* + C(q)z^* = 0, \\ y = Rz^*. \end{cases} \quad (S)$$

$$\dot{y} = R\dot{z}^* = -R[H_0(z^*)\theta^* + C(q)z^*].$$

Estimators $z^\perp$ and θ of $R^\perp$ and θ^* :

$$\begin{cases} \bar{z} = y + z^\perp \\ \dot{z}^\perp + R^\perp(H_0(\bar{z})\theta + C(q)\bar{z}) = \Psi^\perp \\ \dot{\theta} = \Psi^\theta. \end{cases} \quad (1)$$

Similar works: Lepsien et al. [2025](#).

Sketch of proof

$$\begin{cases} \Psi^\perp = \mu [\tilde{H}_0(\theta) + C(q)]^* R \left([\tilde{H}_0(\theta^*) + C(q)] e^z + H_0(\bar{z}) e^\theta \right) \\ \Psi^\theta = \mu [H_0(\bar{z})]^* R \left([\tilde{H}_0(\theta^*) + C(q)] e^z + H_0(\bar{z}) e^\theta \right) \end{cases} \quad (2)$$

$$\begin{aligned} \dot{\Gamma} = & - \left\langle H_0(e^z) \theta^* + H_0(\bar{z}) e^\theta, e^z \right\rangle \\ & - \mu \left\langle [\tilde{H}_0(\theta^*) + C(q)] e^z + H_0(\bar{z}) e^\theta, R H_0(e^z) e^\theta \right\rangle \\ & - \mu \left\| R \left([\tilde{H}_0(\theta^*) + C(q)] e^z + H_0(\bar{z}) e^\theta \right) \right\|^2 \end{aligned}$$

Hypotheses

- continuity (and maybe coercivity) of the operators involved
- smallness of initial parameter error $\theta_0 - \theta^\star$
- Persistence of excitation: $\exists \tau > 0, \exists \varepsilon > 0, \forall T > 0,$

$$\int_T^{T+\tau} \left\| R([\tilde{H}_0(\theta^\star) + C(q)]e^z + H_0(\bar{z})e^\theta) \right\|^2 \geq \varepsilon \|e^\theta\|^2. \quad (3)$$

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What about the control?

$$\exists \textcolor{red}{q}, \exists \tau_{\textcolor{red}{q}} > 0, \exists \varepsilon_{\textcolor{red}{q}} > 0, \forall T > 0,$$

$$\int_T^{T+\tau} \left\| R([\tilde{H}_0(\theta^*) + C(\textcolor{red}{q})] e_{\textcolor{red}{q}}^z + H_0(\bar{z}_{\textcolor{red}{q}}) e_{\textcolor{red}{q}}^\theta) \right\|^2 \geq \varepsilon \|e_{\textcolor{red}{q}}^\theta\|^2. \quad (4)$$

How could one optimise the control?

- maximisation of information $\left\| R([\tilde{H}_0(\theta^*) + C(q)] e^z + H_0(\bar{z}) e^\theta) \right\|^2$ over an acquisition period $[0, T]$?
- open- or closed-loop?

Conclusion & outlook

We have:

- a nice general framework
- an online method that should work

Now we need:

- prove convergence, despite additional noises
- to design efficient and robust controls
- to test it numerically

Outlook on future works:

- one derivative in time seems efficient, could we make use of multiple?
- what about other systems?

Merci pour votre attention !



Baumeister, J, W Scondo, MA Demetriou, and IG Rosen (1997).

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Lepsien, Arthur, Philipp Kügler, and Alexander Schaum (2025). “On the Structural Analysis and Optimal Input Design for Joint State and Parameter Estimation”. In: *IEEE Access* 13, pp. 211944–211959.